

IConMoE: Modeling Intents of Misinformation using Concept Activation Vector-based Mixture of Experts

Atul Kumar Singh
Fujitsu Research of India Pvt. Ltd.
Bengaluru, India
atul.singh@fujitsu.com

Rahul Mishra
Fujitsu Research of India Pvt. Ltd.
Bengaluru, India
Rahul.N.Mishra@fujitsu.com

Abstract

Understanding the intent behind misinformation helps reveal distinct patterns and underlying motives, enabling more accurate and efficient detection. However, existing approaches often struggle to effectively model the interplay of multiple coexisting intents within claims, and typically require additional LLM inferences to infer intent for each sample. In this paper, we introduce an intent-informed concept activation vector-based mixture-of-experts framework, iConMoE, which models latent intents in an unsupervised manner and predicts the veracity of claims in an end-to-end fashion, without the need for additional LLM inferences. Evaluated on three benchmark datasets, our method consistently outperforms both baseline and state-of-the-art models. We also conduct comprehensive ablation studies and error analyses to highlight the contributions of individual components and identify key challenges. In addition, we provide a qualitative analysis of the learned intents across samples to offer deeper insights into the model's reasoning.

CCS Concepts

• **Information systems**; • **Computing methodologies** → *Causal reasoning and diagnostics*; **Natural language processing**;

Keywords

Misinformation detection, Concept Activation Vectors, Mixture-of-Experts, Intent modeling

ACM Reference Format:

Atul Kumar Singh and Rahul Mishra. 2026. IConMoE: Modeling Intents of Misinformation using Concept Activation Vector-based Mixture of Experts. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792582>

1 Introduction

Verifying the veracity of online claims and detecting misinformation typically requires modeling the context surrounding a given claim. This often requires auxiliary sources, such as corroborating evidence from the Web [25, 29], social media propagation patterns [2, 24], or user interactions [16]. However, these external signals are not always available, reliable, or scalable. In contrast, leveraging

 Claim Example:		
Hillary Clinton says: "The crowd (in Las Vegas) fled at the sound of gunshots. Imagine the deaths if the shooter had a silencer. "		
✓ Ground Truth: Satirical exaggeration highlighting gun control debate and political manipulation, not a literal threat.		
Aspect	Single intent model	Multi intent model
 Predicted Intent(s)	Political Manipulation	Political Manipulation + Satire/Parody
 Analysis	Focuses solely on the political aspect and misses the satirical tone. As a result, it may misclassify the claim as a misleading or dangerous literal threat .	Processes the claim as a layered, satirical political statement by jointly modeling both intents, allowing deep interpretation of the underlying communication strategy.
 Contrast	Failure Mode : Ignores nuanced communication strategies (e.g., irony, sarcasm).	Benefit : Enables nuanced modeling of layered communication often used in politically charged misinformation.

Figure 1: Illustration of Single- vs. Multi-Intent Detection in Misinformation Classification.

intrinsic features of misinformation, particularly its underlying intent, offers a promising alternative [38]. Analyzing the underlying intent of misinformation facilitates the identification of systematic patterns and strategic drivers, thereby improving the accuracy and efficiency of automated misinformation detection without relying on user engagement or social dissemination cues. However, prior work largely fails to capture the interaction of multiple coexisting intents within a single claim, instead adopting a simplistic single-intent assumption. Furthermore, these methods [38, 40] often depend on repeated and computationally expensive LLM inferences to infer intent, making them impractical for deployment in scalable real-world misinformation detection systems.

The key challenges in modeling and utilizing intents associated with claims are fivefold: 1) *Unsupervised Inference*: Intents must be inferred in an unsupervised manner, as manually annotating intents for each claim is neither scalable nor cost-effective. 2) *Dynamic Categorization*: Intent categories should be determined dynamically in an end-to-end fashion to support real-world deployment scenarios. 3) *Efficiency Constraints*: The approach should avoid relying on multiple expensive LLM inferences to determine intents. 4) *Multi-Intent Modeling*: Instead of assuming a single intent per claim, the system should identify an appropriate mix of intents to enable more nuanced and context-aware interpretations. 5) *Category Delineation*: There must be a clear separation between selected and omitted intent categories to ensure interpretability and control.



This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.
WWW '26, Dubai, United Arab Emirates
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792582>

In this paper, we aim to address the aforementioned challenges by proposing a method that learns relevant misinformation intents in an unsupervised manner, without relying on large language model forward passes during training or inference. Our approach also models the interplay of multiple coexisting intents within a claim to capture deeper nuances underlying misinformation objectives. We introduce IConMoE, an intent-informed, concept activation vector (CAV)-based mixture-of-experts framework designed for end-to-end claim veracity prediction. IConMoE captures a latent mixture of user intents in an unsupervised manner. Concept activation vectors for each intent category are precomputed offline using user-provided examples, which obviates the need for expensive LLM inferences. Importantly, IConMoE is a plug-and-play module that can be seamlessly integrated into any pretrained model or LLM backbone to inject intent-aware reasoning.

We employ cross-attention to learn the alignment between potential intent CAVs and the claim representations, while a mixture-of-experts mechanism models the latent combination of relevant intents. To ensure stable training and effective expert routing, we incorporate a KL divergence loss. Furthermore, we apply a contrastive loss on the intent-informed CAV representations to promote semantic disentanglement and enhance the distinction between intent classes.

Unlike prior baselines such as DM-INTER, which require repeated intent-specific LLM forward passes with a time complexity of $O(K \times N \times D)$, our proposed IConMoE performs a single-pass encoding followed by lightweight intent alignment, reducing the overall complexity to $O(N \times (D + K))$, where K is the number of intents, N is the number of input samples, and D is the number of decoder layers. This makes IConMoE substantially more efficient for multi-intent inference at scale.

The key contributions of this work are as follows: 1) We introduce, for the first time to our knowledge, a method for modeling intents behind misinformation using precomputed Concept Activation Vectors (CAVs), thereby eliminating the need for expensive LLM inferences. 2) We propose IConMoE, a novel framework that leverages a CAV-based mixture-of-experts (MoE) architecture to effectively learn the appropriate mixture of coexisting intents within claims. 3) We provide a comprehensive evaluation of our model across three benchmark datasets, demonstrating consistent improvements over relevant baselines and state-of-the-art methods. 4) We conduct extensive ablation studies to systematically analyze and reveal the impact of each design choice and component, highlighting their contributions to the overall effectiveness of the architecture in determining claim veracity.

2 Related Work

Our approach draws on and unifies ideas from four research areas: misinformation detection, intent-aware analysis, concept-based interpretability, and mixture-of-experts architectures with routing and contrastive objectives.

Misinformation Detection. Misinformation detection has been a central research focus due to its societal impact. The early methods combined linguistic characteristics, propagation patterns, and metadata cues using traditional machine learning [2, 3, 22, 24]. With

the advent of deep learning, CNNs and RNNs were adopted to capture local and sequential dependencies [4, 25, 31]. More recently, pretrained transformers such as BERT [7], RoBERTa [20], and DeBERTa [13] have become the backbone of misinformation detection, enabling fine-grained contextual modeling [12]. Hybrid approaches incorporating multimodal or social signals [15, 34, 37] further improved robustness. Despite these advances, most approaches treat misinformation as a binary veracity classification task, without explicitly capturing the intent that drives misinformation creation.

Intent-Aware Misinformation Detection. Intent plays a key role in shaping misinformation, as motives such as political manipulation, economic gain, or satire yield distinct patterns of spread and reception. Prior work modeled user behavior and diffusion dynamics to approximate intent [23, 37, 45], but these are largely indirect. More structured approaches rely on curated intent taxonomies [1, 41], which provide useful frameworks but require costly annotation. Wang et al. (2024) [38] introduced intent embeddings from human-curated categories to enrich semantic features, while Wang et al. (2025) [40] extended this with graph-based intent semantics co-modeling. Related work in stance detection and propaganda classification [31] addresses latent objectives behind the text but lacks explicit disentanglement of semantic and intent signals. In contrast, **IConMoE** learns mixtures of intents in an unsupervised manner.

Concept-Based Interpretability. Concept Activation Vectors (CAVs), introduced by Kim et al. [17], provide post-hoc explanations by quantifying the sensitivity of a model to human-defined semantic concepts. Extensions such as LG-CAV [14] enhance concept learning in low-data regimes, primarily for vision-language tasks. In NLP, CAV are used to sensitize models towards novel unseen concepts not covered in training for hate speech detection [11]. Beyond interpretability, steering methods including Contrastive Activation Addition [28], Concept Bottleneck Models [18] and GCAV [44] show how concept supervision can guide predictions, yet no prior work has explored using mixtures of CAVs for intent modeling in misinformation detection.

Mixture-of-Experts and Routing Mechanisms. Mixture-of-Experts (MoE) architectures enable conditional computation by activating only a subset of specialized expert modules per input, scaling models efficiently [10, 33]. Early routing mechanisms such as top- k gating suffered from load imbalance and expert collapse [10]. Alternatives like expert choice and dynamic routing alleviate these issues, often using auxiliary losses to encourage expert diversity [8, 19]. MoEs have also been applied in domain adaptation and multitask learning by specializing experts by topic or style [33].

Notably, **IConMoE** aligns gating and experts with interpretable intent-specific CAVs and employs contrastive and KL-divergence regularizations to maintain fidelity to human-understandable semantic concepts.

3 Methodology

In this section, we present IConMoE, our intent-informed Concept Activation Vector Mixture-of-Experts framework. We first motivate the role of intent representations in misinformation detection, then outline the key components: (i) encoding intents as Concept Activation Vectors (CAVs), (ii) Mixture of Intent-Expert routing, and

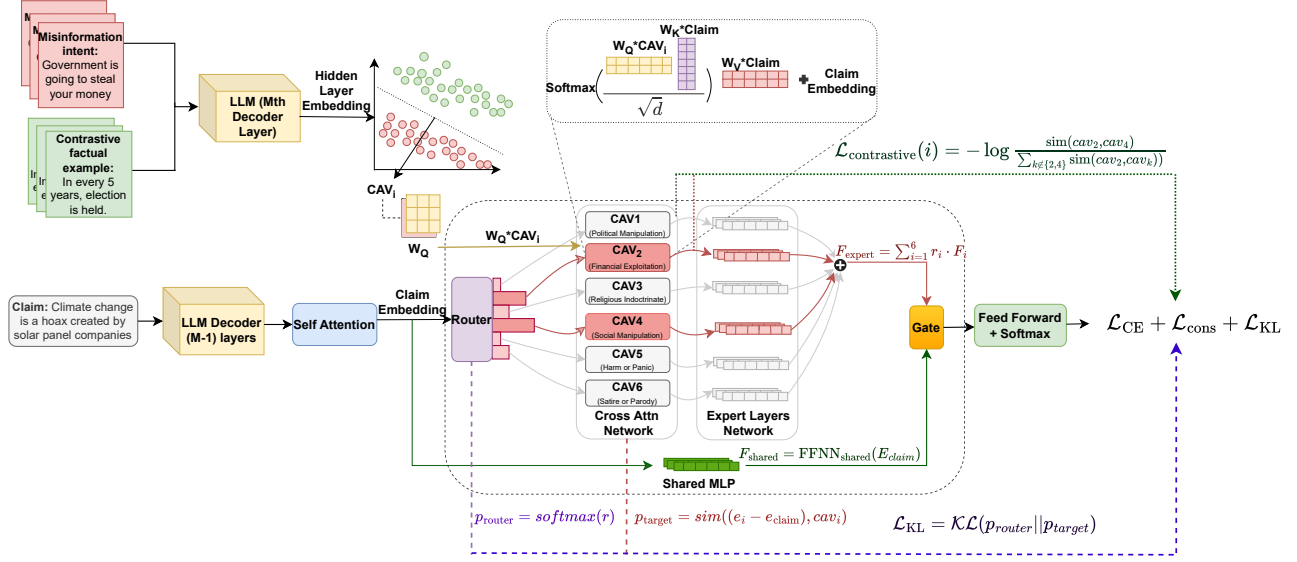


Figure 2: Overview of the IConMoE architecture. Intent-specific Concept Activation Vectors (CAVs) guide cross-attention to project claim embeddings into semantically aligned intent subspaces. The mixture-of-experts module, governed by a learned router, combines intent-specialized and shared experts through gated fusion. Auxiliary KL divergence and contrastive losses enhance routing consistency, semantic disentanglement, and interpretability across expert pathways.

(iii) joint KL and contrastive loss for veracity prediction. Figure 2 illustrates the overall architecture.

3.1 Intent Dataset Construction

We define a representative taxonomy of six misinformation intents based on most frequently observed intent types in widely used misinformation corpora (Politifact and Snopes) and real-world fake news cases, denoted as $\mathcal{I} = \{\text{political manipulation, financial exploitation, ideological/religious indoctrination, social/psychological manipulation, malicious harm or panic, satire/parody misinterpreted}\}$. Analyses of these datasets show that these intents collectively cover the majority of misinformation narratives, spanning political, social and economic dimensions. While additional intents can be modeled easily in our framework, we intentionally adopt a compact taxonomy to demonstrate that even a small set of interpretable intents can result in strong performance gain.

We first construct a small intent dataset using Gemini[6]. For **Positive samples** (x^+) for intent i , we prompt model to output misinformation sample containing that particular intent and for **Negative samples** (x^-), we prompt model to output factual claims matched in topic and style but not expressing intent. This yields a balanced dataset of 200 samples per intent, which are used for computing linearly separable concept vectors. Since CAVs require only approximate linear separability, prior work [17] demonstrates that such compact concept datasets are sufficient to effectively compute CAVs, highlighting the efficiency of the CAV-based approach. Refer to A.7 for intent vs non-intent based examples.

3.2 Computing Intent based Concept Activation Vectors

To represent intent categories in the embedding subspace, we extract Concept Activation Vectors (CAVs), which define semantic directions in the hidden layer embedding space of a pretrained language model. Each CAV is captured by the discriminative axis that separates examples aligned with a particular intent from its contrastive, non-intent counterparts.

Formally, let $CAV = cav_1, cav_2, \dots, cav_k$ denote the set of intent vectors which defines interpretable semantic directions aligned with intent categories, with each $cav_k \in \mathbb{R}^H$ corresponding to one of K intent categories. We encode intent (x^+) and contrastive (x^-) samples, extracting embeddings $\hat{h}^+, \hat{h}^- \in \mathbb{R}^H$ after self-attention layer. Then, we train a linear classifier to separate $\hat{h}^+$ from $\hat{h}^-$, giving weight vector w_i . Normalizing the w_i yields the intent concept activation vector: $cav_i = w_i / \|w_i\|$

Each vector cav_k is orthogonal to the separating hyperplane, ensuring that the vector captures the essence of the concept itself, rather than artifacts of the classifier. CAVs are computed once using offline concept datasets and kept fixed during training to maintain interpretability and reduce overfitting.

3.3 CAV-Guided Cross-Attention

To integrate Concept Activation Vectors (CAVs) into the model, we design a CAV-guided cross-attention mechanism that allows each intent-specific CAV to probe claim embeddings and highlight sequence regions aligned with that intent. For six intent categories, let $cav_i \in \mathbb{R}^H$ denote the CAV for intent i , where H is the embedding dimensionality of the pretrained language model and $E_{claim} \in \mathbb{R}^{S \times H}$

be the claim embeddings where S is the sequence length, and H the embedding dimension after the self-attention layer. We project them into query, key, and value spaces:

$$Q_{cav_i} = W_Q cav_i, \quad K_{claim} = W_K E_{claim}, \quad V_{claim} = W_V E_{claim}, \quad (1)$$

where $W_Q, W_K, W_V \in \mathbb{R}^{H \times H}$ are learnable projection matrices. Using the CAV-conditioned queries, we compute the scaled dot-product attention for each intent category:

$$\text{CAV-CrossAttn}_i = \text{softmax} \left(\frac{Q_{cav_i} K_{claim}^\top}{\sqrt{d_k}} \right) V_{claim}. \quad (2)$$

The use of CAV-based queries Q_{cav_i} ensures that each attention head is directly conditioned on a human-interpretable semantic concept corresponding to specific intent i . To integrate this intent-aware information back into the claim representation, we use a residual connection:

$$E'_{cav} = E_{claim} + \text{CAV-CrossAttn}_i. \quad (3)$$

This residual update allows the model to retain the original claim information while enriching it with intent-specific context, which facilitates downstream expert specialization.

3.4 Intent-Specific Mixture of Experts

After aligning the claim embeddings to semantically meaningful subspaces using CAV-guided attention, we pass them through a set of intent-specific expert networks. Experts are computational modules specialized to intents, each expert corresponds directly to one of the six predefined intent categories. Formally, for each expert $i \in \{1, \dots, 6\}$, the transformed representation is computed as:

$$F_i = \text{LayerNorm} \left(E'_{cav} + \text{Expert}_i(E'_{cav}) \right), \quad (4)$$

where $\text{Expert}_i(\cdot)$ is a lightweight two-layer MLP defined as:

$$\text{Expert}_i(h) = W_{i,2} \cdot \text{ReLU}(W_{i,1}h + b_{i,1}) + b_{i,2}, \quad (5)$$

with weights $W_{i,1} \in \mathbb{R}^{d \times d}$, $W_{i,2} \in \mathbb{R}^{d \times d}$, and corresponding biases $b_{i,1}, b_{i,2} \in \mathbb{R}^d$, all learnable parameters unique to expert i . This two-layer design enables richer transformations within each intent subspace while maintaining parameter efficiency and allowing experts to learn more expressive mappings to refine intent-specific signals.

3.5 Router Weighting and Final Aggregation

A central challenge in modeling misinformation is that claims often exhibit multiple overlapping intents rather than belonging to a single mutually exclusive category. To address this, we adopt a soft router mechanism that adaptively allocates attention among experts. Given the claim embedding $E_{claim} \in \mathbb{R}^d$, the router computes a probability vector $\pi(x) \in \mathbb{R}^6$, where each element $\pi_i(x)$ represents the probability of routing through expert i :

$$\pi(x) = \text{softmax}(W_r E_{claim}), \quad (6)$$

with $W_r \in \mathbb{R}^{6 \times d}$ being a trainable routing matrix. This softmax formulation ensures $\sum_i \pi_i(x) = 1$, defining a valid categorical distribution over the six experts.

Each i -th expert $f_i(x)$ transforms the aligned claim representation using intent-specialized computations. The final expert output

is the expectation over all expert outputs under the router distribution:

$$F_{\text{expert}} = \sum_{i=1}^6 \pi_i(x) \cdot f_i(x). \quad (7)$$

This allows smooth flow across intent dimensions, making it well-suited for claims that express mixed or ambiguous intent, allowing modeling of claims with multiple overlapping intents.

3.6 Shared Expert and Gated Combination

We augment the expert module with a shared feedforward network, denoted $\text{FFNN}_{\text{shared}}$, which captures additional intents that are not covered in our defined set of intents (such as popularize, communicate for true claims). While intent-specific experts capture fine-grained distinctions among misinformation strategies, this shared expert serves as 1) a mechanism to have a smooth gradient flow for better stability in training 2) a complementing strategy for the intent-informed experts by capturing undefined intents.

$$\text{Expert}_{\text{shared}} = \text{FFNN}_{\text{shared}}(E_{claim}), \quad (8)$$

where $\text{FFNN}_{\text{shared}}$ is a two-layer MLP with ReLU activation:

$$\text{FFNN}_{\text{shared}}(h) = W_{s,2} \cdot \text{ReLU}(W_{s,1}h + b_{s,1}) + b_{s,2}, \quad (9)$$

with $W_{s,1}, W_{s,2} \in \mathbb{R}^{d \times d}$ and $b_{s,1}, b_{s,2} \in \mathbb{R}$.

Since not all samples benefit equally from intent-enriched representation (from intent experts) versus generic representation (from shared expert), we introduce a gating mechanism to dynamically adjust their contributions. Specifically, a sigmoid gate computes the weight of the shared pathway as a function of the claim embedding:

$$g_{\text{shared}} = \sigma(W_g^T E_{claim} + b_g), \quad (10)$$

where $W_g \in \mathbb{R}^{d \times 1}$, $b_g \in \mathbb{R}^d$, and $\sigma(\cdot)$ is the sigmoid function. The final output representation is a gated combination of the expert and shared outputs:

$$F_{\text{final}} = g_{\text{shared}} \odot \text{Expert}_{\text{intent}} + (1 - g_{\text{shared}}) \odot \text{Expert}_{\text{shared}}, \quad (11)$$

where $\odot$ denotes element-wise multiplication. This gate helps the model to decide for the underlying sample whether to emphasize on intent-enriched representation from intent experts or generic representation from shared expert. Please refer to discussion about shared gate value in Appendix 5.5.

3.7 Router Regularization via KL Divergence

We guide the MoE router with intent-aligned Concept Activation Vectors (CAVs) because no external supervision specifies the correct expert for a claim. Let router logits be $r \in \mathbb{R}^K$ and

$$p_{\text{router}} = \text{softmax}(r), \quad (12)$$

which also encodes the model's belief about the relevance of each expert. For each expert i we compute a stable cosine alignment between the expert contribution $e_i - e_{\text{claim}}$ and the intent CAV cav_i :

$$s_i = \frac{(e_i - e_{\text{claim}}) \cdot cav_i}{\|e_i - e_{\text{claim}}\| \cdot \|cav_i\|}. \quad (13)$$

This cosine similarity quantifies how well the expert's modification aligns with its intended semantic direction. A high score indicates that the expert meaningfully contributes to modeling

the intended intent. The scores are converted into a soft target distribution:

$$p_{\text{target}} = \text{softmax}(s), \quad (14)$$

The router is trained to match its predicted probabilities to the CAV-guided target using KL divergence:

$$\mathcal{L}_{\text{KL}} = p_{\text{router},i} \cdot \log \left(\frac{p_{\text{router},i}}{p_{\text{target},i} + \epsilon} \right), \quad (15)$$

where ϵ is a small constant to ensure numerical stability. This target distribution rewards experts that demonstrably shift the claim’s representation along their designated semantic axis. This encourages the router to place mass on experts whose outputs move the representation along interpretable intent axes.

3.8 Contrastive Loss on CAV-Guided Attention

While Concept Activation Vectors (CAVs) are inherently interpretable, multiple CAV-guided cross-attention heads may converge to overlapping or redundant signals. To mitigate this, we introduce a contrastive separation objective that encourages each CAV-guided cross-attention head to capture a unique semantic direction. Intuitively, the contrastive term pushes expert representations for different intents apart, avoiding entanglement between, say, ‘social-based’ and ‘political’ manipulation vectors.

Let CAV-CrossAttn_i denote the cross-attention context vector produced by the i -th CAV for a given input token position which treats claim as anchor and computes similarity with all the CAVs. For each claim, cross-attention value acts as an alignment score and then we identify the top two most aligned cross-attention outputs as positives (CAV-CrossAttn_i , CAV-CrossAttn_j), while treating the remaining cross-attention outputs as negatives for the contrastive loss. For contrastive objective, we use InfoNCE [27] loss as:

$$\mathcal{L}_{\text{contrastive}}(i) = -\log \frac{\exp(z_i/\tau) + \exp(z_j/\tau)}{\sum_{k \notin \{i,j\}} \exp \text{sim}(z_k/\tau) + \epsilon}, \quad (16)$$

where $z_i = \text{CAV-CrossAttn}_i$, τ is a temperature parameter and ϵ ensures numerical stability. This loss enforces semantic diversity across intent-specific attention heads, ensuring that each CAV captures a distinct and non-redundant axis of intent.

3.9 Final Training Objective

The overall training objective integrates task-level supervision with two auxiliary losses that enforce intent-specific disentanglement and faithful expert routing:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda \cdot \mathcal{L}_{\text{contrastive}} + \beta \cdot \mathcal{L}_{\text{KL}}, \quad (17)$$

where $\mathcal{L}_{\text{task}}$ is the standard cross-entropy loss for predicting claim veracity, and λ, β are scalar hyperparameters that balance the contributions of the auxiliary objectives. This multi-objective formulation stabilizes training, improves generalization to unseen claims, and ensures that the model’s reasoning paths remain transparent and aligned with human-understandable intents. Detailed analysis of hyperparameters (λ, β) present in A.6

4 Experimental Settings

We design our experiments to evaluate the effectiveness of our intent-based framework and analyze the contribution of its components.

4.1 Datasets

We evaluate IConMoE on three widely used misinformation detection benchmarks that collectively cover political, social, and rumor-style content. To emulate real-world generalization, we adopt data splits consistent with prior work. Table 7 summarizes dataset statistics.

PolitiFact [29] and **Snopes** [36] serve as *intent-aligned* datasets, as the misinformation claims they contain align closely with the intent categories modeled in our framework (e.g., political manipulation, social polarization, financial exploitation). PolitiFact originally provides six graded veracity labels—*true*, *mostly true*, *half true*, *mostly false*, *false*, and *pants-on-fire*—which we merge into binary classes: *real* (the first three) and *fake* (the latter three), following common practice. Snopes comprises fact-checked claims curated from the official Snopes fact-checking portal and is annotated with binary veracity labels.

In contrast, **GossipCop** [34] is considered an *intent-diverse* dataset. It contains approximately 11.8K entertainment and rumor-based news articles labeled as *real* or *fake*. For temporal robustness, we follow the standard setup by training on data up to 2017 and testing on claims from 2018. Unlike PolitiFact and Snopes, the claims in GossipCop often center around celebrity rumors or popular culture misinformation, that does not directly correspond to the intent categories defined in our framework.

4.2 Baselines

We benchmark IConMoE against diverse misinformation detection baselines, covering pretrained transformers, event-aware models, graph-based frameworks, and intent-conditioned methods. All are trained under identical splits for fair comparison. Section A.3 contains detailed description of baseline models.

Pretrained Transformers: BERT [7] serve as strong text-only encoders, T5(base/large) [30] is encoder-decoder model evaluated in finetuned setting while LLaMA-3.1 (8B) [9] is evaluated in in-context and finetuned modes.

Event- and Emotion-Aware Networks: EANN [39] learns event-invariant representations via adversarial objectives, BERT-Emo [45] integrates emotion in BERT and ENDEF [46] applies entity-level debiasing to mitigate overfitting.

Multi-Source and Structural Models: MDFEND and GenFEND [26] incorporate domain and multimodal information, while MGIN-AG [43], MSynFD [42] exploit syntactic or graph-based cues for relational reasoning.

Intent-Based Models: DM-Inter [38] models auxiliary intent vectors through self-supervised intervention.

5 Results and Discussion

Our experiments are structured (refer to A.4 for detailed experimental setup) to evaluate IConMoE across intent-aware and base models, compare it against diverse state-of-the-art baselines, and analyze the contributions of KL and contrastive losses, mixture of experts, and concept activation vectors. Additionally, we conduct qualitative router analysis under KL and contrastive loss, explore alternative CAV computation methods, perform layer-wise examination, and assess the hyperparameter sensitivity of the loss functions.

5.1 Performance of IConMoE Across Intent Aware and Base Models

To further establish the effectiveness of **IConMoE**, we evaluate its integration with a range of base architectures, from large decoder-only LLMs (LLaMA-3.1 8B) to smaller and larger encoder-decoder models (T5-base, T5-large). Table 1 reports results on GossipCop, Snopes, and PolitiFact. We compare four settings: (1) zero-shot in-context learning (ICL), (2) intent-enhanced LLMs where the closest intent tokens are concatenated to inputs, (3) standard fine-tuning of base models, (4) DM-Inter[38]

Across all three misinformation datasets, GossipCop, Snopes, and PolitiFact, **IConMoE** delivers consistent and significant gains over baseline methods and prior intent-aware models. DM-Inter is built on top of T5-base, across all datasets, **IConMoE** (T5-base) consistently outperforms DM-Inter. On *GossipCop*, it improves Macro-F1 from 79.8 to 81.5, on *Snopes*, **IConMoE** lifts performance from 64.5 to 67.3 Macro-F1, and on *PolitiFact*, from 61.3 to 66.2. These gains confirm that modeling mixture of intent as structured concept vectors with experts captures finer-grained semantic variation than DM-Inter’s single-intent intervention mechanism. For large decoder-only models, **IConMoE** also yields significant improvements over conventional fine-tuning. On *GossipCop*, it raises Macro-F1 from 46.2 to 65.3 (+19.1 points), while on *Snopes* it climbs from 58.3 to 67.5 (+9.1), and on *PolitiFact* from 55.8 to 63.2 (+7.4).

Key insights from the results are: 1) LLaMA-3.1 8B alone performs weakly on misinformation detection under ICL or fine-tuning, yet with **IConMoE**, it reaches competitive performance which shows **IConMoE** acts as lifeline for weaker baselines. 2) **IConMoE** consistently outperforms prior baseline (DM-Inter) across datasets and model families. This shows that intent mixture modeling and CAV alignment are more effective than direct intent embeddings. 3) Performance gains on both T5-base and T5-large suggest that **IConMoE** scales favorably with model size, but even smaller models benefit substantially. 4) **IConMoE** boosts both encoder-decoder and decoder-only small/large models, validating its general-purpose applicability.

5.2 Comparison with Diverse State-of-the-art Baselines

Table 2 presents a comprehensive comparison of **IConMoE** against a wide range of existing misinformation detection baselines on **PolitiFact** and **GossipCop**.

On **PolitiFact**, **IConMoE** with T5 large achieves the best performance across all metrics, obtaining 0.6874 Macro-F1 and 0.6875 Accuracy. The improvement is particularly clear in class-specific F1-scores: **IConMoE** balances detection across real (0.6809) and fake (0.6939) classes, whereas prior methods (e.g., GenFEND, DM-Inter) tend to skew toward one class. The advantage here is important: **PolitiFact** contains politically charged claims with subtle intent differences, where explicit intent embeddings (DM-Inter) or emotional features (BERT-Emo) alone are insufficient. Since the **PolitiFact** dataset deals primarily with political claims, the intents we define (e.g., political manipulation, social manipulation) are highly relevant. This strong alignment between dataset content and intent categories allows **IConMoE** to capture nuanced intent signals that improve classification.

On **GossipCop**, **IConMoE** achieves strong performance (0.8423 Macro-F1, 0.8870 Accuracy), significantly outperforming most prior baselines such as GenFEND (0.8367) and MSynFD (0.8051). **GossipCop** focuses on celebrity-related gossip which does not align with our predefined intents. Nevertheless, **IConMoE** still shows strong performance. This is because the shared representation layer of our architecture effectively captures undefined intents, allowing the model to generalize to gossip-specific cases that were not explicitly covered by our intent taxonomy.

We note that **INSIDE** [40] reports slightly higher performance on **GossipCop**. However, **IConMoE** outperforms **INSIDE**’s reported results on **PolitiFact**. Direct comparison should be interpreted with caution, as **INSIDE**’s codebase and data splits are not publicly released and differ from both our setup and prior standard partitions. **INSIDE** employs a complex graph-based reasoning module that requires explicit graph construction and additional annotation alignment, leading to higher computational and design overhead.

In contrast, a critical strength of **IConMoE** lies in its adaptability, **IConMoE** functions as a *plug-and-play, model-agnostic module* that integrates seamlessly into any pretrained LLM backbone without supervising external intent signals, handcrafted graphs, or task-specific adaptations.

5.3 Ablation study

To better understand the contribution of each design choice in **IConMoE**, we conduct an extensive ablation analysis on both the **PolitiFact** and **GossipCop** datasets using T5-base. We systematically disable or modify components of the framework to evaluate their impact on performance. Results are shown in Table 3 and Table 8. *Variants Considered*. We evaluate the following ablation settings:

- **w/o KL Regularization**: removes the KL-divergence loss
- **w/o Contrastive Loss**: excludes the contrastive loss
- **Top-2 Routing**: replaces soft intent routing with a hard top-2 gating mechanism.
- **w/o Shared Expert**: removes the shared expert, forcing all features through experts.
- **w/o Intent Experts**: removes the expert-specific MLPs and only takes the cross-attention values.
- **Single Combined Intent**: merges all intents into a generic “misinformation” class, discarding fine-grained intents.

Key Findings. **1) Regularization Matters:** Both KL divergence and contrastive loss are crucial for stabilizing routing and Cross-attention output. Removing either reduces Macro F1 by 2-3 points across datasets. The contrastive loss especially affects **PolitiFact**, showing the importance of separating overlapping intents. **2) Routing Strategy:** Replacing soft routing with top-2 hard routing slightly improves **GossipCop** but degrades **PolitiFact**, suggesting that political misinformation often involves mixtures of intents best modeled by probabilistic routing. **3) Shared Layer Contribution:** Excluding the shared semantic layer consistently harms performance especially model performance on real claim. This supports our hypothesis that shared expert helps to capture the nuances of factual claim. **4) Expert Specialization:** Removing expert MLPs leads to strong drop in the performance. This confirms that modular experts capture distinct misinformation strategies that a single classifier

Table 1: Comparison of IConMoE with fine-tuned and intent-based baselines across GossipCop, Snopes, and PolitiFact. Statistical significance was confirmed using a paired t-test ($p < 0.05$) between IConMoE (T5-l) and baselines.

Model	GossipCop					Snopes					PolitiFact				
	Acc	F1 _M	F1 _R	F1 _F	AUC	Acc	F1 _M	F1 _R	F1 _F	AUC	Acc	F1 _M	F1 _R	F1 _F	AUC
ICL (LLaMA 8B)	59.7	52.6	79.1	34.4	61.6	55.8	54.7	47.8	61.6	57.9	52.5	52.5	51.3	53.7	52.5
Intent-LLM (8B)	58.2	51.1	69.7	32.7	60.2	56.3	55.2	48.4	62.1	58.4	54.2	53.0	60.4	45.5	55.1
Finetuned (Llama)	72.9	46.2	84.1	8.3	64.8	67.3	58.3	39.1	77.6	67.3	56.7	55.8	49.5	62.1	56.1
Finetuned (T5-b)	85.7	79.8	97.5	68.9	85.2	69.8	48.5	15.4	81.6	63.9	59.2	58.1	51.5	64.8	58.5
Finetuned (T5-l)	86.7	81.4	91.3	71.4	86.2	73.0	64.0	46.0	82.0	72.4	64.2	60.4	59.2	61.5	60.4
DM-Inter	85.5	79.8	95.3	69.0	85.0	74.1	64.5	46.4	82.7	72.9	61.4	61.3	63.1	59.4	65.7
IConMoE (Llama)	79.3	65.3	85.6	45.1	74.7	71.8	67.5	55.7	79.3	73.0	63.3	63.2	65.6	60.7	63.7
IConMoE (T5-b)	86.9	81.5	91.6	71.5	86.4	77.3	67.3	49.2	83.5	75.7	66.3	66.2	67.2	65.2	66.3
IConMoE (T5-l)	88.7	84.2	92.5	76.0	88.2	77.8	67.7	49.7	85.7	76.1	68.8	68.7	68.1	69.4	67.9

Table 2: Performance comparison of IConMoE with diverse state-of-the-art misinformation detection systems on PolitiFact and GossipCop datasets.

Method	Politifact			
	Macro F1	Accuracy	F1 (Real)	F1 (Fake)
BERT	60.41	60.83	61.99	60.87
EANN	61.07	61.46	61.59	62.07
MDFEND	61.95	60.27	70.95	52.95
BERT-Emo	62.51	62.72	61.48	63.35
ENDEF	61.89	62.84	63.57	61.29
MGIN-AG	62.28	62.51	63.01	64.94
MSynFD	62.61	62.52	60.02	65.28
GenFEND	63.80	63.83	64.05	63.66
DM-Inter	61.77	61.51	63.20	59.66
IConMoE	68.74	68.75	68.09	69.39

Method	Gossipcop			
	Macro F1	Accuracy	F1 (Real)	F1 (Fake)
BERT	78.75	84.39	89.73	67.81
EANN	79.31	85.18	90.40	68.23
MDFEND	80.10	85.61	90.22	69.60
BERT-Emo	79.12	84.55	89.74	68.49
ENDEF	80.10	85.20	90.20	69.87
MGIN-AG	80.43	85.93	90.74	70.69
MSynFD	80.51	87.02	91.53	70.34
GenFEND	83.67	85.98	89.95	73.02
DM-Inter	79.60	85.24	90.07	68.86
IConMoE	84.23	88.70	92.46	76.00

cannot. 5) **Granularity of Intents**: Collapsing all intents into a single “misinformation” class yields the worst results. This validates our hypothesis: explicit modeling of intent diversity is critical for robust detection.

5.4 Qualitative Analysis of Router Decisions

To further examine the role of auxiliary losses in improving expert selection, we conduct a qualitative evaluation of router outputs. For each input, we obtain the top-2 intent CAVs selected by the

Table 3: Ablation Study: Comparison of IConMoE Variants on Politifact and Gossipcop.

Variant	Politifact			
	Acc	F1 (R)	F1 (F)	Macro F1
IConMoE	66.25	67.21	65.24	66.22
IConMoE w/o KL	63.75	62.98	64.49	63.74
IConMoE w/o Cons	64.58	63.20	65.86	64.53
IConMoE Top2 routing	65.42	65.84	64.98	65.41
IConMoE w/o Shared Expert	65.00	65.85	64.10	64.98
IConMoE w/o Intent Experts	62.50	64.57	60.18	62.37
IConMoE 1-Intent	60.83	61.48	60.17	60.82

router and passed the query to LLM (Gemini 2.5)[6] acting as a semantic judge. We used LLM as a judge instead of human to ensure consistency and avoiding the subjective human bias. Gemini assigns a score of 3 if both CAVs correspond to valid intents, 2 if only the top CAV is correct, 1 if only the second is correct, and 0 if neither is valid. Table 4 presents results on Politifact and Snopes.

Table 4: Qualitative Router Analysis: Average Gemini-assigned correctness scores (0-3) for top-2 intent CAVs.

Model Variant	Politifact	Snopes
IConMoE (T5 base)	1.93	2.08
IConMoE (T5 base) w/o KL	1.71	1.44
IConMoE (T5 base) w/o KL + contrastive	1.61	1.10

We observe that **IConMoE (T5 base)** achieves the highest semantic scores (1.93 and 2.08), indicating that both contrastive and KL regularization jointly guide the router toward intent-consistent expert choices. These findings highlight that the both auxiliary objectives are *complementary*: the contrastive term ensures that intent-aligned CAVs are well separated in embedding space, while the KL term prevents degenerate expert assignments by softly regularizing the routing distribution. Together, they encourage the router to select intents that not only improve downstream accuracy but also match human-interpretable categories, thus improving the performance of IConMoE predictions.

5.5 Significance of shared gate

We observed that the shared gate exhibits two characteristic behaviors. For *fake* claims, the gate consistently assigns higher weights to specialist experts, emphasizing intent-specific reasoning. In contrast, for *real* claims, the gate mostly down-weights these intent-enriched representation and relies on the generic representation from shared expert. This trend is crucial: the shared expert captures intents that lie outside the intent taxonomy. When a specialist expert is spuriously activated by intent-like tokens, the shared expert provides a corrective signal. Formally, the model output $\hat{z}(x) = \text{softmax}(g(x)E(x) + (1 - g(x))S(x))$ thus uses $S(x)$ as a robust fallback representation. In practice this reduces false positives on real claims (where intent signals are weak or absent) and improves calibration by smoothing over overconfident specialist votes.

5.6 Analysis of classifier-free CAV computation

Traditional CAV extraction requires training a linear classifier (e.g., SVM or logistic regression) to separate positive and contrastive samples for each concept and each layer, which can be computationally expensive at scale. To address this, we experiment with a *classifier-free* approach, where each Concept Activation Vector (CAV) is computed as the normalized mean difference between positive and negative intent embeddings: $cav_i = (\bar{h}_i^+ - \bar{h}_i^-) / (\|\bar{h}_i^+ - \bar{h}_i^-\|)$

Table 5: Comparison of fine-tuned, classifier-based, and classifier-free CAV (embedding difference) approaches on Politifact and Gossipcop datasets.

Model	Politifact			
	Acc	F1(R)	F1(F)	Macro F1
Finetune (T5-base)	59.17	51.49	64.75	58.12
IConMoE (CAV-classifier)	66.25	67.21	65.24	66.22
IConMoE (CAV-embedding)	63.75	61.67	65.61	63.64
Model	Gossipcop			
	Acc	F1(R)	F1(F)	Macro F1
Finetune (T5-base)	85.74	90.75	68.89	79.82
IConMoE (CAV-classifier)	86.97	91.43	72.50	81.97
IConMoE (CAV-embedding)	86.55	91.37	69.49	80.43

Table 5 compares standard fine-tuning, classifier-based IConMoE, and the proposed embedding-difference CAV method on both **Politifact** and **Gossipcop**. The results show that the classifier-free variant achieves performance close to the classifier-based CAV, outperforming the finetuned baseline by a significant margin. This formulation maintains the interpretability of the geometric direction, while eliminating the need for separate classifiers. It also ensures consistent scaling across intents and simplifies deployment during multi-intent routing.

5.7 Layer-wise CAV intervention and interpretability study

We injected the IConMoE architecture at different depths of the model to analyze where intent-sensitive representations most effectively influence misinformation detection. We inject our architecture at 1st layer, mid layer and last layer to judge the effectiveness

Table 6: Layer-wise impact of CAV integration in IConMoE (T5-base) on Politifact and Snopes.

Model Variant	Politifact			
	Acc.	F1 (R)	F1 (F)	Macro F1
IConMoE — last layer	66.25	67.21	65.24	66.22
IConMoE — 1st layer	62.92	57.00	67.40	62.20
IConMoE — 6th layer (mid)	63.33	59.63	66.41	63.02
Model Variant	Snopes			
	Acc.	F1 (R)	F1 (F)	Macro F1
IConMoE — last layer	77.25	49.16	85.35	67.25
IConMoE — 1st layer	74.50	50.49	82.83	66.66
IConMoE — 6th layer (mid)	75.25	50.75	83.47	67.11

of IConMoE architecture at different layers. Results in Tables 6 and 9 indicates that adding IConMoE at the final layer consistently yields the best macro-F1 across both datasets and backbones. This suggests that high-level semantic activations near the output layer carry stronger decision cues than lower or mid-level linguistic abstractions.

To quantify where internal prediction shifts occur, we measure the layer l^* at which the difference between real and fake prediction probabilities changes most sharply representing a **decision flip** within the model’s reasoning hierarchy:

$$l^* = \arg \max_l |\delta_l - \delta_{l-1}|, \quad (18)$$

where δ_l denotes the difference between fake and real logits at layer l . Empirically, we observe that l^* predominantly yields a higher value at last layer and occasionally with the 0th layer. This suggests that intent-conditioned representations peak at the output layer where contextual and factual signals merge.

6 Conclusions and Future Prospects

In this paper, we propose IConMoE, an intent-informed Concept Activation Vector (CAV)-based mixture-of-experts model designed to effectively capture intent mixtures underlying misinformation. IConMoE leverages precomputed concept vectors to model potential intents without relying on large language model (LLM) inferences during training or inference, offering a highly efficient and scalable alternative to existing methods. Through evaluations on three widely-used benchmark datasets, IConMoE demonstrates significant performance improvements over both strong baselines and state-of-the-art models, all while maintaining an end-to-end, computationally light framework.

In future work, we plan to investigate the interplay between emotions and intents in determining misinformation veracity. Additionally, we aim to extend our approach to multi-modal misinformation detection, particularly by modeling intent representations in the image modality alongside those in text.

References

- [1] Basant Agarwal, Ajay Agarwal, Priyanka Harjule, and Azizur Rahman. 2023. Understanding the intent behind sharing misinformation on social media. *Journal of Experimental & Theoretical Artificial Intelligence* 35, 4 (2023), 573–587.
- [2] Tian Bian, Xi Xiao, Tingyang Xu, Peilin Zhao, Wenbing Huang, Yu Rong, and Junzhou Huang. 2020. Rumor detection on social media with bi-directional graph convolutional networks. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 34. 549–556.

- [3] Carlos Castillo, Marcelo Mendoza, and Barbara Poblete. 2011. Information credibility on twitter. In *Proceedings of the 20th international conference on World wide web*. 675–684.
- [4] Tong Chen, Xue Li, Hongzhi Yin, and Jun Zhang. 2018. Call attention to rumors: Deep attention based recurrent neural networks for early rumor detection. In *Pacific-Asia conference on knowledge discovery and data mining*. Springer, 40–52.
- [5] Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Yunxuan Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, et al. 2024. Scaling instruction-finetuned language models. *Journal of Machine Learning Research* 25, 70 (2024), 1–53.
- [6] Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Naveen Sachdeva, Inderjit Dhillon, Marcel Blstein, Ori Ram, Dan Zhang, Evan Rosen, et al. 2025. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261* (2025).
- [7] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*. 4171–4186.
- [8] Nan Du, Yanping Huang, Andrew M Dai, Simon Tong, Dmitry Lepikhin, Yuanzhong Xu, Maxim Krikun, Yanqi Zhou, Adams Wei Yu, Orhan Firat, et al. 2022. Glam: Efficient scaling of language models with mixture-of-experts. In *International conference on machine learning*. PMLR, 5547–5569.
- [9] Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. 2024. The llama 3 herd of models. *arXiv e-prints* (2024), arXiv–2407.
- [10] William Fedus, Barret Zoph, and Noam Shazeer. 2022. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research* 23, 120 (2022), 1–39.
- [11] Samarth Garg, Vivek Hruday Kavuri, Gargi Shroff, and Rahul Mishra. 2024. KTCR: Improving Implicit Hate Detection with Knowledge Transfer driven Concept Refinement. *arXiv preprint arXiv:2410.15314* (2024).
- [12] Andreas Hanselowski, Avinash PVS, Benjamin Schiller, Felix Caspelherr, Debanjan Chaudhuri, Christian M Meyer, and Iryna Gurevych. 2018. A retrospective analysis of the fake news challenge stance detection task. *arXiv preprint arXiv:1806.05180* (2018).
- [13] Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. 2020. Deberta: Decoding-enhanced bert with disentangled attention. *arXiv preprint arXiv:2006.03654* (2020).
- [14] Qihan Huang, Jie Song, Mengqi Xue, Haoifei Zhang, Bingde Hu, Huiqiong Wang, Hao Jiang, Xingen Wang, and Mingli Song. 2024. Lg-cav: Train any concept activation vector with language guidance. *Advances in Neural Information Processing Systems* 37 (2024), 39522–39551.
- [15] Dhruv Khattar, Jaipal Singh Goud, Manish Gupta, and Vasudeva Varma. 2019. Mvae: Multimodal variational autoencoder for fake news detection. In *The world wide web conference*. 2915–2921.
- [16] Ling Min Serena Khoo, Hai Leong Chiew, Zhong Qian, and Jing Jiang. 2020. Interpretable Rumor Detection in Microblogs by Attending to User Interactions. *Proceedings of the AAAI Conference on Artificial Intelligence* 34, 05 (Apr. 2020), 8783–8790. doi:10.1609/aaai.v34i05.6405
- [17] Been Kim, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viegas, et al. 2018. Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (tcav). In *International conference on machine learning*. PMLR, 2668–2677.
- [18] Pang Wei Koh, Thao Nguyen, Yew Siang Tang, Stephen Mussmann, Emma Pierson, Been Kim, and Percy Liang. 2020. Concept bottleneck models. In *International conference on machine learning*. PMLR, 5338–5348.
- [19] Dmitry Lepikhin, Hyoukjoong Lee, Yuanzhong Xu, Dehao Chen, Orhan Firat, Yanping Huang, Maxim Krikun, Noam Shazeer, and Zhifeng Chen. 2020. Gshard: Scaling giant models with conditional computation and automatic sharding. *arXiv preprint arXiv:2006.16668* (2020).
- [20] Yinhan Liu, Mylène Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692* (2019).
- [21] Ilya Loshchilov and Frank Hutter. 2017. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101* (2017).
- [22] Jing Ma, Wei Gao, Prasenjit Mitra, Sejeong Kwon, Bernard J Jansen, Kam-Fai Wong, and Meeyoung Cha. 2016. Detecting rumors from microblogs with recurrent neural networks. (2016).
- [23] Euna Mehnaz Khan, Ayush Ram, Bhavtosh Rath, Emily Vraga, and Jaideep Srivastava. 2023. Behavioral Forensics in Social Networks: Identifying Misinformation, Disinformation and Refutation Spreaders Using Machine Learning. *arXiv e-prints* (2023), arXiv–2305.
- [24] Rahul Mishra. 2020. Fake News Detection Using Higher-Order User to User Mutual-Attention Progression in Propagation Paths. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*.
- [25] Rahul Mishra and Vinay Setty. 2019. SADHAN: Hierarchical Attention Networks to Learn Latent Aspect Embeddings for Fake News Detection (ICTIR '19). Association for Computing Machinery, New York, NY, USA, 197–204. doi:10.1145/3341981.3344229
- [26] Qiong Nan, Qiang Sheng, Juan Cao, Beizhe Hu, Danding Wang, and Jintao Li. 2024. Let silence speak: Enhancing fake news detection with generated comments from large language models. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*. 1732–1742.
- [27] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. 2018. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748* (2018).
- [28] Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. 2023. Steering llama 2 via contrastive activation addition. *arXiv preprint arXiv:2312.06681* (2023).
- [29] Kashyap Popat, Subhabrata Mukherjee, Andrew Yates, and Gerhard Weikum. 2018. Declare: Debunking fake news and false claims using evidence-aware deep learning. *arXiv preprint arXiv:1809.06416* (2018).
- [30] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. 2020. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of machine learning research* 21, 140 (2020), 1–67.
- [31] Hannah Rashkin, Eunsol Choi, Jin Yea Jang, Svetlana Volkova, and Yejin Choi. 2017. Truth of varying shades: Analyzing language in fake news and political fact-checking. In *Proceedings of the 2017 conference on empirical methods in natural language processing*. 2931–2937.
- [32] Nils Reimers and Iryna Gurevych. 2019. Sentence-bert: Sentence embeddings using siamese bert-networks. *arXiv preprint arXiv:1908.10084* (2019).
- [33] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziars, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. 2017. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv preprint arXiv:1701.06538* (2017).
- [34] Kai Shu, Amy Sliva, Suhang Wang, Jiliang Tang, and Huan Liu. 2017. Fake news detection on social media: A data mining perspective. *ACM SIGKDD explorations newsletter* 19, 1 (2017), 22–36.
- [35] Gemma Team, Thomas Mesnard, Cassidy Hardin, Robert Dadashi, Surya Bhupatiraju, Shreya Pathak, Laurent Sifre, Morgane Rivière, Mihir Sanjay Kale, Juliette Love, et al. 2024. Gemma: Open models based on gemini research and technology. *arXiv preprint arXiv:2403.08295* (2024).
- [36] Nguyen Vo and Kyumin Lee. 2020. Where are the facts? searching for fact-checked information to alleviate the spread of fake news. *arXiv preprint arXiv:2010.03159* (2020).
- [37] Soroush Vosoughi, Deb Roy, and Sinan Aral. 2018. The spread of true and false news online. *science* 359, 6380 (2018), 1146–1151.
- [38] Bing Wang, Ximing Li, Changchun Li, Bo Fu, Songwen Pei, and Shengsheng Wang. 2024. Why misinformation is created? detecting them by integrating intent features. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*. 2304–2314.
- [39] Yaqing Wang, Fenglong Ma, Zhiwei Jin, Ye Yuan, Guangxu Xun, Kishlay Jha, Lu Su, and Jing Gao. 2018. Eann: Event adversarial neural networks for multi-modal fake news detection. In *Proceedings of the 24th acm sigkdd international conference on knowledge discovery & data mining*. 849–857.
- [40] Zhengjia Wang, Qiang Sheng, Danding Wang, Beizhe Hu, and Juan Cao. 2025. Bridging Thoughts and Words: Graph-Based Intent-Semantic Joint Learning for Fake News Detection. *arXiv preprint arXiv:2509.01660* (2025).
- [41] Thomas D Wilson and Elena Maceviciute. 2022. Information misbehaviour: modelling the motivations for the creation, acceptance and dissemination of misinformation. *Journal of Documentation* 78, 7 (2022), 485–505.
- [42] Liang Xiao, Qi Zhang, Chongyang Shi, Shoujin Wang, Usman Naseem, and Liang Hu. 2024. Msynfd: Multi-hop syntax aware fake news detection. In *Proceedings of the ACM web conference* 2024. 4128–4137.
- [43] Weizhi Xu, Junfei Wu, Qiang Liu, Shu Wu, and Liang Wang. 2022. Evidence-aware fake news detection with graph neural networks. In *Proceedings of the ACM web conference* 2022. 2501–2510.
- [44] Hanyu Zhang, Xiting Wang, Chengao Li, Xiang Ao, and Qing He. 2025. Controlling large language models through concept activation vectors. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 25851–25859.
- [45] Xinyi Zhou, Kai Shu, Vir V Phoha, Huan Liu, and Reza Zafarani. 2022. “this is fake! shared it by mistake”: Assessing the intent of fake news spreaders. In *Proceedings of the ACM web conference* 2022. 3685–3694.
- [46] Yongchun Zhu, Qiang Sheng, Juan Cao, Shuokai Li, Danding Wang, and Fuzhen Zhuang. 2022. Generalizing to the future: Mitigating entity bias in fake news detection. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 2120–2125.

A Appendix

A.1 Dataset statistics

Table containing the dataset statistics of politifact, snopes and gossipcop:

Table 7: Dataset statistics showing the number of *real* and *fake* instances across train, validation, and test splits.

Dataset	Train		Validation		Test	
	Fake	Real	Fake	Real	Fake	Real
GossipCop	2,024	5,039	604	1,774	601	1,758
PolitiFact	1,074	939	136	114	134	118
Snopes	2,343	888	269	131	293	107

A.2 Gossipcop ablation study

Table 8: Ablation Study: Comparison of IConMoe Variants on Politifact and Gossipcop.

Variant	Gossipcop			
	Acc	F1 (R)	F1 (F)	Macro F1
IConMoE	86.97	91.43	72.50	81.97
IConMoE w/o KL	86.89	91.46	71.83	81.64
IConMoE w/o Cons	86.38	91.30	68.68	79.99
IConMoE Top2 routing	86.93	91.56	71.49	81.53
IConMoE w/o Shared	86.47	91.32	69.36	80.34
IConMoE w/o Experts	85.62	90.44	71.00	80.72
IConMoE 1-Intent	85.83	90.48	72.31	81.39

A.3 Detailed description of Baselines

We compare IConMoE against both general-purpose models and specialized MD methods. In particular, DM-Inter is our primary baseline due to its close design goals. Below is a brief description of each:

- BERT [7] : A transformer-based encoder pretrained via masked language modeling and next-sentence prediction which is widely used in text classification.
- T5 (base/large) [30] : A text-to-text transformer that can be fine-tuned for classification tasks. We include both T5-base and T5-large to test capacity effects.
- LLaMA-3.1 (8B)[9] : large-scale LLM evaluated in in-context learning (ICL), intent-augmented (LLM-Int), and fine-tuning settings. To ensure computational efficiency, we fine-tune only its final classification layer.
- DM-Inter[38] : A self-supervised intent-intervention framework where generator is trained to output binary intent predictions, which are sharpened and integrated with news embeddings for veracity judgment.
- EANN [39] : Learns event-invariant representations using adversarial training, aiming to reduce event-based biases in news misinformation.

- MDFEND[26] : Employs a mixture-of-experts paradigm to account for domain variation in news to maintain cross-domain robustness.
- BERT-Emo [45] : Enhances a BERT-based model with emotional features to capture affective signals useful in misinformation.
- ENDEF [46] : Introduces entity-based debiasing to reduce the influence of highly frequent entities in news that may skew detection.
- MGIN-AG [43] : Combines syntactic dependency graphs with BERT, leveraging graph structure to augment representation for detection.
- MSynFD[42] : Builds multi-hop syntactic dependency graphs over news tokens to integrate syntax-aware signals for improved detection.
- GenFEND[26] : Generates synthetic user comment feedback via LLM prompts, aggregates them, and uses them to enhance misinfo detection.

A.4 Implementation Details

All experiments were implemented using the Hugging Face Transformers library with AdamW [21] as the optimizer and a cosine annealing schedule with 100 warm-up steps. We trained for 5 epochs with a batch size of 8, a learning rate of 1e-6, and weight decay set to 0.01. Mixed precision (FP16) was disabled to avoid training instabilities, though BF16 can be adopted on supported hardware. We evaluated our approach across four strong pretrained backbones: FLAN-T5-Base and FLAN-T5-Large [5], Google/Gemma-7B [35], and Meta-LLaMA-3.1-8B [9]. For intent-aware modeling, we first obtained intent embeddings using the all-MiniLM-L6-v2 encoder [32], and then determined the closest intent for each claim by computing cosine similarity between the claim and intent vectors. All experiments were conducted on NVIDIA H100 GPUs with at 90GB memory.

A.5 Layer-wise CAV intervention and interpretability study

Layer-wise impact of IConMoE integration in LLaMA-8B on Politifact and Snopes. This shows integrating integrating IConMoE in last layer shows most advantage.

Table 9: Layer-wise impact of CAV integration in IConMoE (LLaMA-8B) on Politifact and Snopes.

Model Variant	Politifact			
	Acc.	F1 (R)	F1 (F)	Macro F1
IConMoE — last layer	63.33	65.62	60.71	63.16
IConMoE — 1st layer	62.92	55.72	68.10	61.91
IConMoE — 15th layer (mid)	61.67	43.31	71.00	57.11

Model Variant	Snopes			
	Acc.	F1 (R)	F1 (F)	Macro F1
IConMoE — last layer	71.75	55.69	79.27	67.48
IConMoE — 1st layer	69.25	39.41	79.40	59.40
IConMoE — 15th layer (mid)	71.00	10.77	82.69	46.73

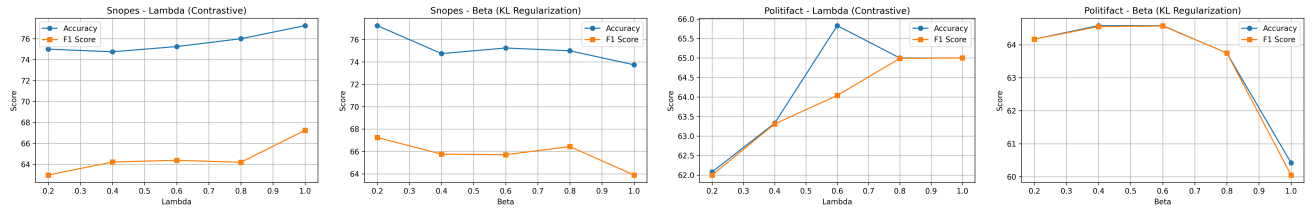


Figure 3: Effect of contrastive loss (λ) and KL regularization (β) on Snopes and Politifact datasets.

A.6 Hyperparameter analysis

We further investigate the sensitivity of IConMoE to the weighting of auxiliary objectives, namely the contrastive consistency loss (λ) and the KL regularization loss (β). Figures 3 report the effect of varying λ and β in the range $[0.2, 1.0]$ on Snopes and Politifact datasets. We observe that as λ approaches 1.0, both accuracy and macro F1 consistently improve on Snopes, reaching the highest scores when the contrastive term dominates. A similar upward trend is visible for Politifact, where performance peaks around $\lambda = 0.6$ before plateauing. These results indicate that the contrastive term is crucial for stabilizing expert separation by pushing apart conflicting intents. In contrast, increasing β has a diminishing effect on performance. On Snopes and politifact, both accuracy and macro F1 steadily decline as β grows, indicating that excessive regularization over-constrains expert routing.

A.7 Intent dataset example

Factual Claims (No Intent)	Misinformation Claims (With Intent)
Taylor Swift has one of the largest fanbases among singers.	Taylor Swift is helping fans become financially free with this simple trick. (Financial exploitation)
Face masks help protect people from viruses.	Wearing face masks causes lung cancer, stop using them immediately. (Harm/Panic)
Taking breaks during work can reduce stress and improve productivity.	Government introduces 'Nap Hour' between 2 and 3 PM in all workplaces. (Political Manipulation)

Table 10: Samples from intent dataset for CAV computation

A.8 Prompts for intent example generation and analysis

To construct intent-specific positive and negative samples, we use the following prompt to generate synthetic claim pairs with a large language model. Each pair consists of two short claims on the same fictional topic and in a similar writing style.

Generate two short fictional claims on the same made-up topic and in the same style.

Intent: {political manipulation | financial exploitation | ideological/religious indoctrination | social/psychological

manipulation | malicious harm or panic | satire/parody misinterpreted}

Positive (x^+): A fictional false claim that clearly expresses the given intent.

Negative (x^-): A fictional neutral or factual claim on the same topic that does not express the intent.

Do not use real people, organizations, places, or events. Label each output as Positive (x^+) or Negative (x^-).

A.9 Error analysis

A.10 CAV generated using other LLMs

To evaluate the robustness of IConMoE to the choice of LLM used for intent dataset construction, we generate Concept Activation Vectors (CAVs) using intent samples produced by ChatGPT instead of Gemini. As shown in Table 12, IConMoE maintains comparable performance across both CAV sources, with only a marginal drop when using ChatGPT-based CAVs. This indicates that our framework is not sensitive to a specific LLM and benefits primarily from the intent-informed CAV formulation rather than the generator model. Gemini-2.5 produced most consistent and least stylistically biased examples.

A.11 Varying number of intent

We study the impact of intent granularity by varying the number of intent categories k . Results in Table 13 show that performance degrades gradually as k decreases, with the full six-intent setting achieving the best results. This suggests that a compact yet sufficiently expressive intent taxonomy is important for capturing diverse misinformation strategies, while fewer intents limit the model’s ability to disentangle semantic intent signals.

A.12 Ethics statement

This work uses large language models (LLMs) solely to generate synthetic misinformation examples for offline concept extraction. These samples are used only to compute intent-specific Concept Activation Vectors (CAVs) and are never included in model training, evaluation, deployment, or public release.

All generation is constrained by strict safety and content moderation settings, ensuring that samples remain abstract, non-operational, and do not reference real events, provide instructions, or target individuals or groups. Generated data are stored locally and deleted after concept computation.

The proposed IConMoE framework does not require LLM inference during training or inference, minimizing risks of unintended

Table 11: Error analysis of IConMoE illustrating how intent combinations and router dynamics influence prediction outcomes. Each column shows a representative claim with gold label, detected intents, and reasoning behaviour.

Aspect	Claim 1: Refugees from neighboring state of the country are being paid to protest in capital city rallies.	Claim 2: The government banned diesel cars to force citizens into buying expensive electric vehicles owned by foreign companies.	Claim 3: Barack Obama says Medicare and Medicaid are the single biggest drivers of the federal deficit and the federal debt by a huge margin.
Gold Label	Fake	Fake	Real
Top Intents (Router Selection)	Political manipulation, Social manipulation	Satire/Parody, Political manipulation, Financial exploitation	Political manipulation, Financial exploitation
Model Behaviour and Internal Dynamics	The router activates high gating weights for <i>political</i> and <i>social manipulation</i> , allowing specialist experts to capture the coordinated hostility narrative. The expert attention maps align with causal verbs like “paid” and “protest,” enabling confident discrimination of synthetic fear appeals.	Although the claim is correctly classified as fake, the router over-weights the <i>Satire/Parody</i> expert despite the lack of humor markers. The joint contribution from political and financial manipulation dominates, yielding the correct prediction through cross-intent reinforcement.	The router misfires by over-activating manipulation intents, mapping fiscal terms (<i>deficit, debt</i>) to exploitative framing. Lack of evidential grounding causes the shared expert’s corrective signal to be downweighted, leading to a false positive.
Analysis and Insight	Captures fine-grained intent alignment and contextual hostility cues. Strength lies in disentangling overlapping manipulation signals that amplify fake framing.	Shows robustness to partial intent mismatch, leveraging mixture gating to stabilize cross-intent dependencies. Suggests the model’s capacity for redundancy in expert reasoning.	Highlights failure case of over-activation bias, where factual but politically charged content mimics manipulative tone. Points to need for better calibration between shared and intent-specific experts.

Table 12: Comparison of fine-tuned, gemini-based dataset CAV and ChatGPT based dataset CAV

Model	Politifact			
	Acc	F1(R)	F1(F)	Macro F1
Finetune (T5-base)	59.17	51.49	64.75	58.12
IConMoE (CAV-Gemini)	66.25	67.21	65.24	66.22
IConMoE (CAV-ChatGPT)	65.00	65.05	65.08	64.99

Table 13: Comparison of IConMoE by varying the number of intents

Model	Politifact			
	Acc	F1(R)	F1(F)	Macro F1
IConMoE(k=6)	66.25	67.21	65.24	66.22
IConMoE(k=5)	65.00	64.90	64.87	64.88
IConMoE(k=4)	64.17	64.10	64.12	64.10

content generation in deployment. The method is designed exclusively for improving misinformation detection and aligns with responsible AI practices and ACM ethical guidelines.